CHEBYSHEV SPECTRAL METHODS FOR HYDROGEN TRANSPORT SIMULATIONS IN METALS

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Abstract. We develop a Chebyshev pseudospectral collocation solver for the McNabb–Foster hydrogen transport equations and benchmark it against FESTIM, a FEniCS-based finite element code, on three problems: constant-temperature permeation with an analytical reference, multi-temperature gas-driven permeation where FESTIM's Newton solver diverges, and thermo-desorption spectroscopy with a Levenberg–Marquardt inverse problem. The Chebyshev solver achieves geometric spatial convergence, delivers matched-accuracy solutions with 8–10× speedups, and maintains convergence in stiff regimes where the finite element solver fails. We find that backward Euler's first-order temporal error is the binding constraint on accuracy at moderate resolution, and that higher-order time integration is needed to unlock the full potential of spectral spatial discretization.

1. Introduction. Nuclear fusion has the potential to provide a virtually limitless supply of clean energy, and significant international effort is now directed toward realizing this goal through devices such as ITER and DEMO [11]. One of the central challenges in the design of these reactors is the management of tritium, the radioactive hydrogen isotope that fuels the deuterium-tritium reaction. Tritium retention in plasma-facing materials poses both a safety concern, because regulatory limits cap the in-vessel inventory, and a fuel-economy concern, because any tritium trapped inside the walls is unavailable for the plasma [16, 18]. For this reason, the ability to accurately predict how hydrogen isotopes diffuse through, become trapped in, and are released from structural metals is a major part of the fusion materials research program [3]. However, much of the relevant material data for fusion-relevant metals is still poorly characterized, and the trapping energies, trap densities, and diffusion activation energies that govern hydrogen transport in tungsten and other candidate structural materials are the subject of active investigation [8, 15]. This is why fast, accurate hydrogen transport codes are needed: not just to simulate known physics, but to serve as the forward model inside inverse problems that extract unknown material properties from experimental measurements.

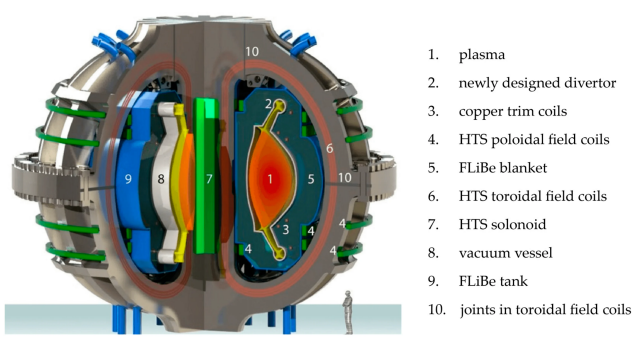
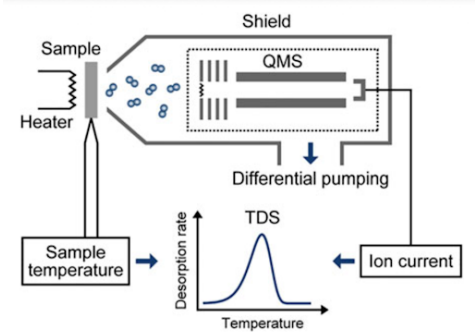
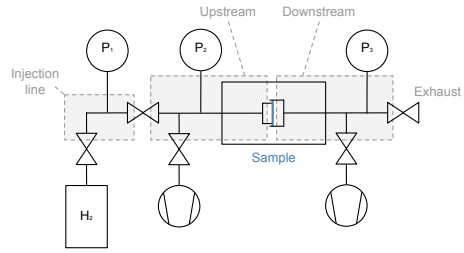


Fig. 1.1: Cross-section of the ARC fusion reactor concept. The structural materials surrounding the FLiBe blanket (5) must withstand permeation and trapping of tritium bred during operation.

Two experimental techniques are the primary diagnostic tools for this extraction. Thermo-desorption spectroscopy (TDS) implants hydrogen into a sample at



(a) Thermo-desorption spectroscopy (TDS). A pre-loaded sample is heated on a linear ramp while a quadrupole mass spectrometer records the desorption flux. The shape of the resulting spectrum encodes trap densities and binding energies.



(b) Gas-driven permeation (GDP). Hydrogen gas is applied at controlled pressure to the upstream face of a foil, and the downstream flux is measured by a pressure gauge. Repeating at several temperatures maps out diffusivity and solubility.

Fig. 1.2: The two experimental techniques used to characterize hydrogen transport in metals.

low temperature, then ramps the temperature and records the desorption flux as a function of time, and the shape of the resulting spectrum encodes the number, density, and binding energy of each trap population [10]. Gas-driven permeation (GDP) applies a controlled hydrogen partial pressure to one face of a foil and measures the downstream flux that emerges on the opposite side, and repeating the experiment at several temperatures maps out the diffusivity and solubility Arrhenius coefficients [9]. In both cases, extracting material properties requires an inverse problem: a forward solver simulates the experiment for a candidate parameter set, and an optimization loop adjusts the parameters until the simulated and measured signals agree. The forward solver is therefore the computational bottleneck of the entire analysis pipeline, and its speed and accuracy directly determine the feasibility of parametric studies, uncertainty quantification, and large-scale fitting campaigns.

The current state-of-the-art tool for these simulations is FESTIM (Finite Element Simulation of Tritium in Materials) [6, 5, 7], which discretizes the McNabb–Foster coupled diffusion-trapping equations [14] using continuous Galerkin finite elements on the FEniCSx platform, with options for P1 (linear), P2 (quadratic), and P3 (cubic) spatial elements, and advances in time with an implicit backward Euler scheme. While FESTIM is accurate and flexible, its spatial discretization converges only algebraically under mesh refinement: $O(h^2)$ for P1, $O(h^3)$ for P2, and $O(h^4)$ for P3, where h is the element size. In practice, this means that resolving the steep concentration gradients that appear near material surfaces and boundary condition transitions requires very fine meshes, particularly in GDP configurations where the diffusivity changes by orders of magnitude between temperature steps. Chebyshev spectral collocation methods, by contrast, offer a fundamentally different convergence rate [19, 1, 2]. For problems with smooth solutions, spectral methods achieve geometric (exponential) convergence, meaning that the error decreases as $O(\rho^{-N})$ for some $\rho > 1$ rather than

as a polynomial power of $1/N$. Furthermore, Chebyshev–Gauss–Lobatto nodes naturally cluster near domain boundaries with a spacing that scales as $O(N^{-2})$ at the endpoints, which is precisely where the sharpest concentration gradients appear in hydrogen transport problems. This combination of rapid convergence and favorable node distribution is why I believe spectral collocation is a strong candidate for accelerating the forward solves that dominate the cost of TDS and GDP analysis pipelines.

In this work, I develop a one-dimensional Chebyshev pseudospectral collocation solver for the McNabb–Foster equations and benchmark it against FESTIM on three representative problems: constant-temperature gas-driven permeation with an analytical reference solution, multi-temperature GDP where FESTIM's Newton solver diverges at deep temperature steps, and TDS with a Levenberg–Marquardt inverse problem. I demonstrate that the Chebyshev solver achieves geometric spatial convergence, delivers matched-accuracy solutions at a fraction of the wall time with speedups of 8–10×, and maintains Newton convergence in regimes where the finite element solver fails. The remainder of the paper is organized as follows: Section 2 presents the governing equations and boundary conditions, Section 3 develops the Chebyshev spectral collocation method and verifies its geometric convergence on a simple permeation benchmark, Section 4 applies the solver to multi-temperature gas-driven permeation where FESTIM's Newton solver diverges, Section 5 addresses thermo-desorption spectroscopy with a multi-domain decomposition and a Levenberg–Marquardt inverse problem, and Section 6 summarizes the findings and discusses future directions.

2. Governing Equations.

2.1. McNabb–Foster model. The McNabb–Foster equations [14] describe the coupled transport of mobile and trapped hydrogen in a metal lattice. The mobile concentration $c_m(x,t)$ obeys a diffusion equation with source and trapping–detrapping coupling, while each trap population $c_{t,i}(x,t)$ evolves according to local kinetics:

$$\frac{\partial c_m}{\partial t} = \frac{\partial}{\partial x}\left(D(T)\,\frac{\partial c_m}{\partial x}\right) + S(x,t) - \sum_i \frac{\partial c_{t,i}}{\partial t}, \tag{2.1}$$

$$\frac{\partial c_{t,i}}{\partial t} = \nu_{m,i}(T)\,c_m\big(n_i - c_{t,i}\big) - \nu_{r,i}(T)\,c_{t,i}. \tag{2.2}$$

Here $D(T)$ is the temperature-dependent diffusivity, $S(x,t)$ is a volumetric source term (e.g., ion implantation), n_i is the density of trap sites of type i, and $\nu_{m,i}$ and $\nu_{r,i}$ are the trapping and detrapping rates, respectively. All temperature-dependent coefficients follow Arrhenius laws:

$$D(T) = D_0\,e^{-E_D/k_BT}, \qquad \nu_{m,i}(T) = \nu^0_{m,i}\,e^{-E_{m,i}/k_BT}, \qquad \nu_{r,i}(T) = \nu^0_{r,i}\,e^{-E_{r,i}/k_BT}, \tag{2.3}$$

where k_B is the Boltzmann constant and E_D, $E_{m,i}$, $E_{r,i}$ are the respective activation energies. The exponential dependence on temperature is what makes the system stiff: a modest change in T can shift the diffusivity by orders of magnitude, producing steep boundary layers and rapid transients that challenge both spatial and temporal discretizations.

2.2. Boundary conditions. Two types of boundary conditions appear in the experiments we consider. For TDS, both surfaces of the foil are held at zero mobile concentration ($c_m = 0$), representing a vacuum boundary from which hydrogen escapes freely. For GDP, the upstream surface satisfies Sieverts' law [17],

$$c_m\big|_{x=0} = K_S(T)\,\sqrt{P(t)}, \qquad K_S(T) = K_{S,0}\,e^{-E_{K_S}/k_BT}, \tag{2.4}$$

where $P(t)$ is the applied hydrogen partial pressure and K_S is the Sieverts solubility, while the downstream surface is held at $c_m = 0$. In both cases, the trapped concentrations have no boundary conditions of their own; they are determined pointwise by the local ODE (2.2).

3. Chebyshev Spectral Collocation Method.

3.1. Chebyshev polynomials and collocation points. The Chebyshev polynomials $T_n(x) = \cos\bigl(n \cos^{-1} x\bigr)$ form an orthogonal basis on $[-1, 1]$ with respect to the weight $(1-x^2)^{-1/2}$. The Chebyshev–Gauss–Lobatto (CGL) nodes are the extrema of $T_N(x)$ together with the endpoints,

$$x_j = \cos\left(\frac{j\pi}{N}\right), \qquad j = 0, 1, \ldots, N, \tag{3.1}$$

which cluster near $x = \pm 1$ with a minimum spacing of $O(N^{-2})$ at the boundaries and a maximum spacing of $O(N^{-1})$ in the interior [19]. This boundary clustering is a natural fit for hydrogen transport problems, where the steepest concentration gradients occur at the material surfaces. To map from the reference interval $[-1, 1]$ to a physical domain $[0, L]$, we apply the affine transformation $\hat{x} = L(1 - x)/2$, which maps $x_0 = 1$ to $\hat{x} = 0$ and $x_N = -1$ to $\hat{x} = L$.

3.2. Differentiation matrices. The key object in Chebyshev collocation is the $(N + 1) \times (N + 1)$ differentiation matrix D_N, whose entries are defined so that the matrix-vector product $D_N \mathbf{u}$ gives the values of the derivative of the interpolating polynomial at the collocation points. Following Trefethen [19], the entries of D_N on the reference interval $[-1, 1]$ are

$$(D_N)_{ij} = \begin{cases} \dfrac{c_i}{c_j}\,\dfrac{(-1)^{i+j}}{x_i - x_j}, & i \neq j, \\ -\dfrac{x_j}{2(1 - x_j^2)}, & 1 \leq i = j \leq N - 1, \\ \dfrac{2N^2 + 1}{6}, & i = j = 0, \\ -\dfrac{2N^2 + 1}{6}, & i = j = N, \end{cases} \tag{3.2}$$

where $c_0 = c_N = 2$ and $c_j = 1$ for $1 \leq j \leq N - 1$. The second-derivative matrix is obtained by squaring: $D_N^{(2)} = D_N^2$. On the physical domain $[0, L]$, the chain rule introduces a scaling factor of $2/L$ per derivative, so the physical first- and second-derivative matrices are $(2/L)\, D_N$ and $(2/L)^2\, D_N^2$, respectively.

If $u(x)$ is analytic in a Bernstein ellipse of semi-axis sum $\rho > 1$ in the complex plane, the interpolation error at CGL nodes satisfies $\|u - p_N\|_\infty = O(\rho^{-N})$, i.e., the error decreases geometrically in N [19, 1]. For C^∞ functions that are not entire, convergence is superalgebraic, and for functions with finite Sobolev regularity H^s, it degrades to algebraic $O(N^{-s})$. In contrast, P1 finite elements converge at the fixed rate $O(h^2)$ regardless of solution smoothness.

3.3. Spatial discretization of the McNabb–Foster system. Let $\mathbf{c}_m = [c_m(x_0), \ldots, c_m(x_N)]^T$ denote the vector of mobile concentrations at the collocation points, and similarly $\mathbf{c}_{t,i}$ for each trapped species. Replacing the spatial derivatives

in (2.1) with the differentiation matrix and evaluating the trapping kinetics pointwise yields the semi-discrete system

$$\frac{d\mathbf{c}_m}{dt} = D(T)\, D_N^{(2)}\, \mathbf{c}_m + \mathbf{S}(t) - \sum_i \frac{d\mathbf{c}_{t,i}}{dt}, \tag{3.3}$$

$$\frac{d\mathbf{c}_{t,i}}{dt} = \nu_{m,i}(T)\, \mathbf{c}_m \odot \big(n_i\, \mathbf{1} - \mathbf{c}_{t,i}\big) - \nu_{r,i}(T)\, \mathbf{c}_{t,i}, \tag{3.4}$$

where $\odot$ denotes the Hadamard (elementwise) product and $\mathbf{S}(t)$ is the source vector evaluated at the nodes. Dirichlet boundary conditions are incorporated by replacing the first and last rows of the diffusion equation with the prescribed values $c_m(x_0, t)$ and $c_m(x_N, t)$ — the standard row-replacement technique for collocation methods [19].

3.4. Time integration. The stiffness introduced by the Arrhenius kinetics (2.3) requires an L-stable implicit time integrator. We use backward Euler (BE) as the default scheme, matching FESTIM's time integrator so that any difference in accuracy between the two solvers is attributable to the spatial discretization alone. Each BE step requires solving a nonlinear system $\mathbf{F}(\mathbf{y}^{n+1}) = \mathbf{0}$, where $\mathbf{y} = [\mathbf{c}_m; \mathbf{c}_{t,1}; \ldots; \mathbf{c}_{t,I}]$ stacks all unknowns. We solve this with a damped Newton iteration using the analytically assembled Jacobian $J = \partial\mathbf{F}/\partial\mathbf{y}$, which has a block structure: the mobile–mobile block contains the dense diffusion operator $D_N^{(2)}$, the mobile–trap and trap–mobile blocks carry the trapping coupling terms, and the trap–trap blocks are diagonal. The time step is controlled adaptively: a growth factor of 1.1 is applied after successful steps, a cutback factor of 0.5 after Newton failures, and milestone times (e.g., the start of the TDS ramp) force the step to land exactly at physically significant transitions.

BE is first-order accurate in time ($O(\Delta t)$), so at a fixed step size cap the temporal error sets a floor below which refining the spatial discretization yields no further improvement. We measure this floor explicitly by comparing BE solutions at Δt and $\Delta t/2$ at high spatial resolution. For the constant-temperature permeation problem at $\Delta t_{\max} = 0.05\,\mathrm{s}$, the BE floor is approximately 9.4×10^{-4} in relative L^2 norm. To demonstrate that the spectral spatial convergence continues below this floor, we also present results with a variable-order BDF integrator (orders 1–5) from SciPy [20], which reduces the temporal floor by roughly a factor of five at comparable wall time.

3.5. Verification: constant-temperature permeation. All benchmarks in this work are run in Python on a single core (Apple M-series), with NumPy/SciPy for the Chebyshev solver and FEniCSx for FESTIM. The Chebyshev implementation mirrors FESTIM's public API class-for-class, so the two solvers differ only in the spatial discretization. For FESTIM, we report the Newton-solve-only wall time (the accumulated cost of the Newton iterations across all backward Euler steps) rather than the total `run()` time, because profiling shows that roughly 60% of FESTIM's per-step overhead is consumed by FFCx JIT cache directory scans during `SurfaceFlux` post-processing, which is independent of mesh resolution. Error is measured as the relative L^2 norm on a common observation time grid, $\|J_{\mathrm{test}} - J_{\mathrm{ref}}\|_2 / \|J_{\mathrm{ref}}\|_2$.

To verify the geometric convergence predicted by the theory above, we consider isothermal gas-driven permeation through a $L = 3 \times 10^{-4}\,\mathrm{m}$ foil at $T = 500\,\mathrm{K}$ with an upstream pressure of $P = 100\,\mathrm{Pa}$ and no trapping. This problem admits a closed-form

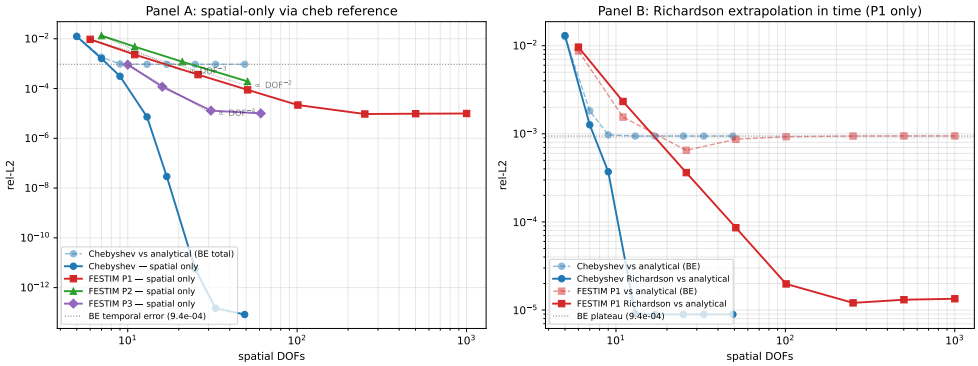


Fig. 3.1: Verification on constant-temperature permeation. Left (Panel A): spatial-only error isolated by measuring both methods against a high-resolution Chebyshev BE solve at the same Δt, so the shared temporal error cancels. Chebyshev converges geometrically; FESTIM P1, P2, and P3 follow slopes -2, -3, and -4 respectively. Right (Panel B): Richardson extrapolation in time drops the temporal floor by a factor of ~ 5.

Fourier series solution for the downstream flux [4],

$$J(t) = \frac{\sqrt{P}\, D\, K_S}{L}\left(1 + 2\sum_{n=1}^{\infty}(-1)^n \exp\left(-\frac{n^2\pi^2 D\, t}{L^2}\right)\right), \tag{3.5}$$

and so serves as an analytical verification benchmark where the spatial discretization error can be measured against truth directly. Both solvers use the same backward Euler controller with $\Delta t_{\max} = 0.05\,\mathrm{s}$ (2000 steps over $t \in [0, 100]$ s), so any difference in relative L^2 error above the temporal floor is purely spatial.

To isolate the spatial discretization error from the temporal floor, Figure 3.1 (Panel A) replaces the analytical reference with a high-resolution Chebyshev BE solve at the same Δt. Since test and reference share the same time integrator, the temporal error cancels, revealing the pure spatial convergence of each method. The Chebyshev curve falls geometrically until it reaches a cross-solver noise floor at approximately 10^{-5}, confirming the geometric convergence predicted by spectral approximation theory. FESTIM P1, P2, and P3 follow their expected algebraic slopes of -2, -3, and -4. Panel B shows that Richardson extrapolation in time ($u_{\mathrm{Rich}} = 2u(\Delta t/2) - u(\Delta t)$) cancels the leading $O(\Delta t)$ backward Euler term, dropping the temporal floor to approximately 2×10^{-4}. This observation is a central finding: backward Euler's first-order temporal error sets a floor that neither spatial method can penetrate, and any effort to improve accuracy beyond the $\sim 10^{-3}$ level must address the time integration regardless of the spatial discretization.

4. Gas-Driven Permeation.

4.1. Configuration. We consider a multi-temperature GDP experiment through a $L = 1\,\mathrm{mm}$ tungsten foil with two traps ($n_1 = 1.3 \times 10^{-3}$, $n_2 = 4 \times 10^{-4}$), cycling the temperature downward in 100 K steps from 700 K and back up. Each temperature segment consists of a constant-pressure run at $P = 1 \times 10^5$ Pa followed by a sinusoidal ramp to $P = 0$ over 20 min (a bake phase that evacuates mobile hydrogen before the

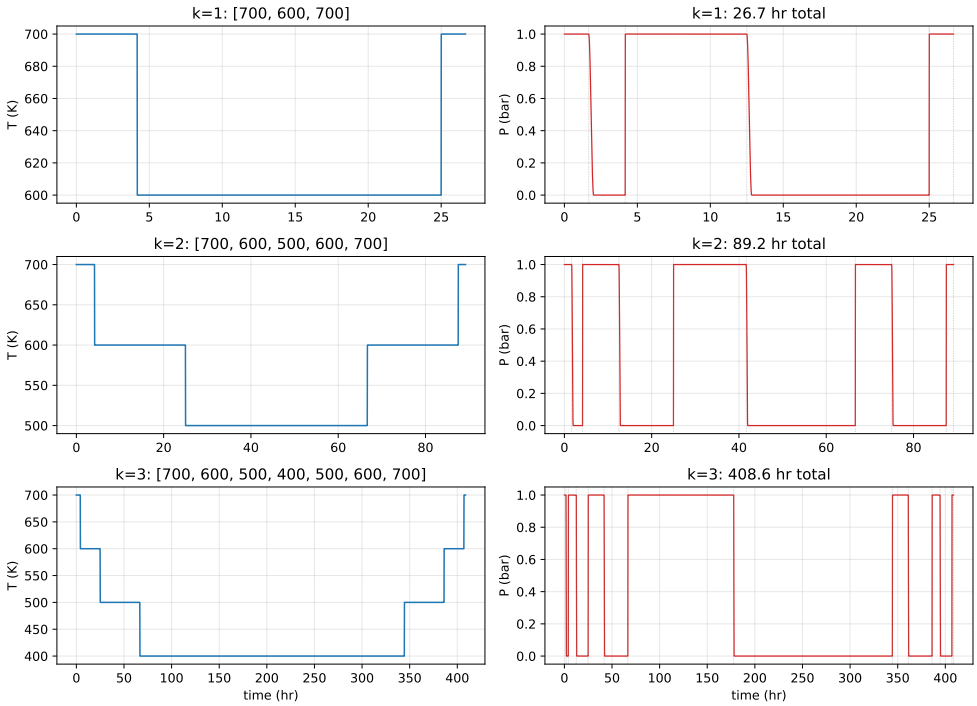


Fig. 4.1: Temperature and pressure schedules for the three multi-temperature GDP protocols ($k = 1, 2, 3$). The $k = 3$ schedule reaches 400 K and spans approximately 409 hours of simulated time.

next temperature step). We consider three schedules of increasing severity: $k = 1$ (700 $\to$ 600 $\to$ 700 K), $k = 2$ (700 $\to$ 600 $\to$ 500 $\to$ 600 $\to$ 700 K), and $k = 3$ (700 $\to$ 600 $\to$ 500 $\to$ 400 $\to$ 500 $\to$ 600 $\to$ 700 K). The $k = 3$ schedule, which reaches 400 K where $D(T)$ is roughly 10^3 times smaller than at 700 K, is known to cause FESTIM's Newton solver to diverge when the pressure boundary condition is reintroduced at higher temperatures after bake phases. The timelag method, which extracts material transport parameters from the transient shape of the permeation curve at each temperature, depends on the solver being able to complete all segments of the cycling protocol; solver divergence at any stage invalidates the entire analysis.

4.2. Results. Figure 4.1 shows the three temperature–pressure cycling protocols. Since no analytical solution exists, we construct a reference by running the Chebyshev solver at $N + 1 = 65$ with $\Delta t_{\max} = 12.5$ s. A same-Δt reference at $N+1 = 65$ with $\Delta t_{\max} = 200$ s isolates the pure spatial error by temporal cancellation. Both methods are swept at $\Delta t_{\max} = 200$ s: Chebyshev at $N + 1 \in \{9, 13, 17, 25, 33\}$ and FESTIM P1 at $n_{\text{cells}} \in \{25, 50, 75, 100, 150\}$ on a graded mesh. A relative L^2 above 0.2 is flagged as silent divergence.

For the mild $k = 1$ schedule, Chebyshev's spatial-only error falls geometrically with N, while FESTIM's error plateaus at a structural offset that is largely independent of mesh density. The flat FESTIM error likely originates from the one-sided gradient extraction used by FESTIM's `SurfaceFlux` post-processing or from dis-

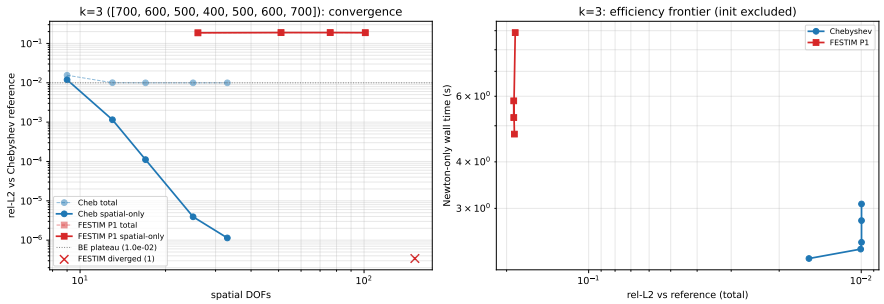


Fig. 4.2: Convergence and efficiency for the severe $k = 3$ schedule (700 $\to$ 400 $\to$ 700 K). Left: relative L^2 error vs. DOFs (bold = spatial-only, dashed = total). Right: Newton-only wall time vs. total error. FESTIM P1 configurations silently diverge (red crosses), while all Chebyshev configurations converge.

cretization artifacts in the reaction coupling. FESTIM's Newton-only wall time is also roughly constant across the tested mesh range (25–150 cells), because at these small DOF counts the actual linear algebra is negligible and the accumulated per-call PETSc solver overhead dominates. For the severe $k = 3$ schedule, however, the situation changes qualitatively. As shown in Figure 4.2, FESTIM's Newton solver fails on several mesh configurations, either explicitly (runtime errors) or silently (relative L^2 error above the 0.2 quality gate). The Chebyshev solver, by contrast, converges on all tested configurations for all three schedules.

This robustness advantage has a clear physical explanation. When the Sieverts boundary condition is reapplied at a higher temperature after a bake phase, both $K_S(T)\sqrt{P}$ and $D(T)$ jump sharply, producing a steep boundary layer near $x = 0$. On a P1 mesh, the boundary layer may span fewer cells than the Newton solver needs to converge, and the iteration either diverges outright or produces a solution that is too smooth to be physically meaningful. The CGL node clustering, however, places $O(N/\pi)$ nodes near each endpoint, providing the resolution the Newton solver needs without requiring a proportionally large total DOF count. For the timelag method, where the solver must complete all temperature segments to produce a valid measurement, this robustness is not a convenience but a requirement for the analysis to proceed at all.

5. Thermo-Desorption Spectroscopy.

5.1. Configuration. In the TDS experiment, a $L = 20\,\mu\text{m}$ tungsten foil is exposed to a Gaussian ion flux of $2.5 \times 10^{19}\,\text{H/m}^2/\text{s}$ centered at $x = 4.5\,\text{nm}$ with a standard deviation of $\sigma = 2.5\,\text{nm}$ for $t \in [0, 400]$ s at $T = 300\,\text{K}$. After a 50 s rest period, a linear temperature ramp begins at $t = 450\,\text{s}$ with a heating rate of $6\,\text{K/s}$. The desorption flux is defined as the sum of the outward diffusive fluxes at both surfaces, $J(t) = -D(T)\,\partial c_m/\partial x\big|_{x=0} + D(T)\,\partial c_m/\partial x\big|_{x=L}$. The true trap parameters are $n_1 = 1.3 \times 10^{-3}$, $n_2 = 4 \times 10^{-4}$ (site fractions), $E_{p,1} = 0.87\,\text{eV}$, and $E_{p,2} = 1.0\,\text{eV}$. The tungsten has two McNabb–Foster traps.

This problem is challenging for both methods because the Gaussian source has a full width at half maximum of approximately 5.9 nm, which is four orders of magnitude smaller than the domain length. A plain Chebyshev–Gauss–Lobatto grid clusters nodes at both endpoints, but the source sits near only one end, so half the spectral

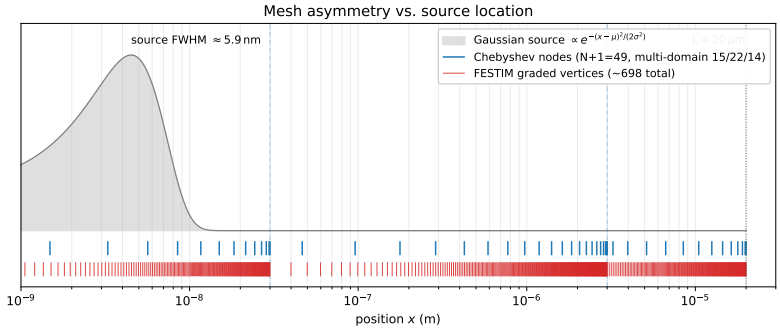


Fig. 5.1: Multi-domain mesh strategy for TDS. The Gaussian source (grey fill, log-x scale) has a FWHM of 5.9 nm inside a 20 µm domain. Chebyshev nodes ($N+1=49$, three blocks at 30 nm / 3 μm / L) and FESTIM graded vertices (∼700 total) are shown.

budget is wasted. Uniform-cell FEM has the complementary problem: to place even ten cells inside the source FWHM would require on the order of 10^5 cells across the full domain.

5.2. Multi-domain spectral elements. The scale separation in the TDS problem motivates decomposing the domain into three subdomains aligned with the source-to-bulk geometry, using the same breakpoints that FESTIM's graded mesh employs: $[0, 30\,\text{nm}]$, $[30\,\text{nm}, 3\,\mu\text{m}]$, and $[3\,\mu\text{m}, L]$. Each subdomain carries its own CGL grid, and the global polynomial budget N is split in a $2:3:2$ ratio between the blocks, matching the vertex-count proportions of the FESTIM graded mesh. Adjacent blocks share their interface node, so the global solution is C^0 continuous across the breakpoints. Flux continuity at the interfaces is enforced by replacing the interface rows in the assembled global system with the condition $D(T)\,\partial c_m/\partial x\big|_{\text{left}} = D(T)\,\partial c_m/\partial x\big|_{\text{right}}$, evaluated using the one-sided differentiation matrices of the two adjacent blocks. The resulting multi-domain scheme has a total of $N+1$ degrees of freedom and retains the geometric convergence of the single-domain method within each block.

5.3. Forward convergence and inverse problem. To handle the four-order-of-magnitude scale separation between the source width and the domain length, both solvers use the three-block decomposition shown in Figure 5.1. The Chebyshev solver splits its polynomial budget $2:3:2$ across the blocks; FESTIM uses a graded mesh with approximately 200/300/200 vertices in the three regions at scale 1.0.

Since no analytical solution exists, we construct a spectrally-converged reference at $N+1=193$. The self-consistency check between $N+1=193$ and $N+1=129$ yields a relative L^2 difference of 1.3×10^{-3}, an order of magnitude below our 10^{-2} target, so the reference is treated as truth.

Figure 5.2 shows the convergence study. The Chebyshev error decreases quickly from 9.9×10^{-1} at $N+1=9$ to 1.3×10^{-3} at $N+1=129$, but is quickly snuffed out by backward Euler eror. FESTIM P1 also converges fairly quickly to the backward Euler error but at a slower rate. At the 10^{-2} target, the smallest qualifying Chebyshev configuration is $N+1=49$ at 0.73 s, while the smallest qualifying FESTIM configuration is mesh scale 0.05 (∼33 vertices) at 5.81 s Newton-only.

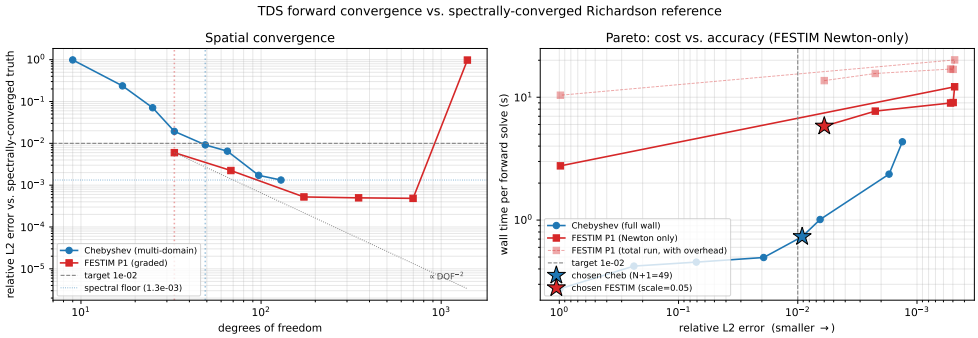


Fig. 5.2: TDS forward convergence. Left: relative L^2 error vs. DOFs. Chebyshev converges geometrically; FESTIM P1 follows slope ≈ -2. Right: wall time vs. error. At the 10^{-2} target, Chebyshev lands at $N+1 = 49$ (0.73 s), FESTIM at scale 0.05 ($\sim$33 vertices, 5.81 s Newton-only) — a per-call speedup of $\sim 8\times$.

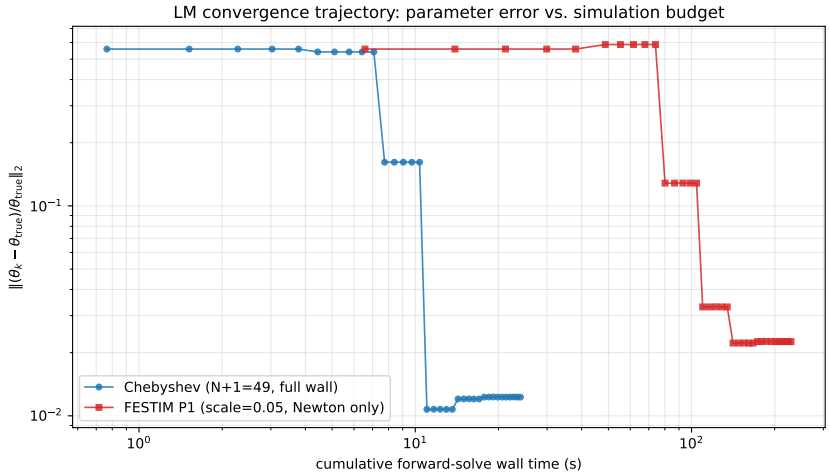


Fig. 5.3: LM convergence trajectory. Parameter error vs. cumulative forward-solve time. Both methods converge in 35 evaluations. Chebyshev completes in 24 s; FESTIM requires 229 s Newton-only — a 9.5$\times$ speedup.

For the inverse problem, we generate synthetic TDS data by evaluating the Chebyshev forward solver at $N+1 = 193$ and adding 2% Gaussian noise, representative of realistic TDS measurement uncertainty. A Levenberg–Marquardt optimizer [12, 13] recovers the four trap parameters $(n_1, n_2, E_{p,1}, E_{p,2})$ from the noisy spectrum. Both methods use their matched-accuracy meshes: Chebyshev at $N+1 = 49$ and FESTIM at scale 0.05.

Figure 5.3 plots the LM convergence trajectory: the relative parameter error $\|(\theta_k - \theta_{\text{true}})/\theta_{\text{true}}\|_2$ as a function of the cumulative forward-solve wall time. Both methods converge in 35 forward evaluations and recover the true parameters to within the noise floor, which is expected given that the 2% measurement noise dominates

the forward-model error at both matched-accuracy meshes. However, the Chebyshev curve sits consistently to the left of the FESTIM curve, reaching any given parameter accuracy with a smaller simulation budget. The cumulative simulation times are 24.0 s for Chebyshev (full wall) and 229.1 s for FESTIM (Newton-only), a 9.5× speedup that compounds over the optimization loop. Because TDS analysis pipelines routinely run hundreds to thousands of such inversions for parametric studies, Bayesian uncertainty quantification, or fitting across sample databases, this per-loop speedup transfers directly into a proportional reduction in the total analysis wall time.

6. Conclusion. We have developed a Chebyshev pseudospectral collocation solver for the one-dimensional McNabb–Foster hydrogen transport equations and benchmarked it against FESTIM on three problems spanning two common experimental configurations in fusion materials characterization. On constant-temperature permeation with an analytical reference, the Chebyshev solver reaches a matched-accuracy target of 2×10^{-3} with 7 collocation nodes in 0.18 s, compared to 11 FESTIM P1 vertices in 16.6 s Newton-only time. On multi-temperature GDP, the Chebyshev solver converges on all configurations including the severe $k = 3$ protocol where FESTIM's Newton solver diverges at deep temperature steps. On TDS with a Levenberg–Marquardt inverse problem, the Chebyshev forward solver achieves a 9.5× cumulative speedup inside the optimization loop (24 s vs. 229 s).

However, if the benefits of Chebyshev collocation are to be fully realized within FESTIM's broader ecosystem, the time integration must be upgraded. Backward Euler's first-order temporal error sets a floor that neither spatial method can penetrate, and this floor dominates the total error budget at the moderate resolutions where the spectral advantage is most pronounced. Higher-order implicit methods — variable-order BDF, implicit Runge–Kutta, or exponential integrators — would lower this floor and allow the geometric spatial convergence to express itself over a wider range of accuracies. The Chebyshev spatial operator, which produces dense but small differentiation matrices, is well suited to coupling with such methods. Extending the solver to multiple dimensions via tensor-product or spectral-element formulations, and integrating it with FESTIM as an alternative spatial backend, are natural directions that could bring these benefits to the broader fusion materials modeling community.

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